SEMESTER VI

XXXX: GENERATIVE AI AND LARGE LANGUAGE MODELS

Objective: *This course aims to introducing the concept of computer organization. In particular, it focuses on basic hardware architectural issues that affect the nature and performance of software.*

Credits: 03 **L–T–P-J: 3–0–0-0**

Module No.	Content	Teaching Hours
I	**Foundations of Generative AI:**Introduction to Generative AI, Generative Models: VAEs, GANs, Autoregressive Models, Transformer Architecture: Encoder-Decoder, Self-Attention, Evolution of LLMs: BERT, GPT, T5, LLaMA, Mixtral, Fine-tuning Methods: Transfer Learning, Prompt Tuning, LoRA, Ethics, Bias, and Responsible AI Practices, Evaluation Metrics: BLEU, ROUGE, Perplexity	**20**
II	**Applied LLMs and GenAI Systems:** Foundation Models and LLM Architectures, Prompt Engineering and In-Context Learning, LangChain, Agents, Tools, and Chains, RAG (Retrieval-Augmented Generation) with Vector Databases, Real-world Use Cases: Chatbots, Document Summarization, Coding Assistants, Open Source LLMs: Deployment with Ollama, LLaMA2, Mistral, LLMOps and Deployment Pipelines	**20**

Text Books:

- Generative AI with Python and TensorFlow2: Create images, text, and music with VAEs, GANs, LSTMs, Transformer models", Joseph Babcock and Raghav Bali,2024
- "Generative AI for everyone: Understanding the essentials and applications of this breakthrough technology". Altaf Rehmani.
- Neural Networks and Deep Learning: A Textbook by Charu C. Aggarwal.

Reference Books:

- Generative Adversarial Networks" by Ian Goodfellow et al.
- Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville.
- Generative Adversarial Networks Cookbook: Over 100 recipes to build generative models using Python, TensorFlow, and Keras" by Josh Kalin.
- Generative AI in Software Development: Beyond the Limitations of Traditional Coding" Jesse Sprinter, 2024..

Outcome: After completion of the course, the student will be able to:

- CO1: Understand the basics of generative AI and LLMs.
- CO2: Demonstrate the principle modelling concept involved in gen AI and LLMs
- CO3: Implement and optimise GANs and VAEs for image generation, including training, fine-tuning, and advanced techniques.
- CO4: Applications of Gen AI and LLMs in real-world scenarios
- CO5: To design solutions for solving practical life problems using generative AI.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO3/PSO1
CO2	PO1,PO3/PSO1
CO3	PO2,PO3,PO5/PSO2
CO4	PO2,PO3,PO4/PSO1,PSO3
CO5	PO2,PO3,PO4/PSO2

XXXX: GENERATIVE AI AND LARGE LANGUAGE MODELS LAB

Objective: *To equip students with theoretical and practical knowledge of Generative AI and LLMs for real-world applications.*

Credits:01 **L-T-P-J:0-0-2-0**

Module No.	Content	Teaching Hours
I& II	• Implement a Variational Autoencoder (VAE) using PyTorch or TensorFlow to generate synthetic handwritten digits from the MNIST dataset. • Design and train a Generative Adversarial Network (GAN) on MNIST to generate new handwritten digit images. • Build a mini transformer model from scratch to perform English-to-French translation using attention mechanisms. • Fine-tune a pre-trained BERT model for sentiment analysis using open datasets and the Hugging Face Transformers library. • Generate text using open-source GPT models like GPT2 or Mistral and perform prompt engineering experiments. • Build a chatbot using LangChain and Streamlit that answers questions from PDF files using local embedding and LLMs. • Create a RAG pipeline using FAISS to retrieve relevant document chunks and respond with an open-source LLM like LLaMA2 or Mistral. • Implement a code generation tool using open-source CodeLLaMA or StarCoder models hosted locally for autocompleting Python code. • Develop an agent using LangChain that can perform reasoning tasks with tools like calculator, search, and memory using local models. • Package and deploy a GenAI microservice (e.g., chatbot or summarizer) using FastAPI and Docker with open-source LLMs.	24

Reference Books:

- Generative Adversarial Networks Cookbook: Over 100 recipes to build generative models using Python, TensorFlow, and Keras" by Josh Kalin.
- Generative AI in Software Development: Beyond the Limitations of Traditional Coding" Jesse Sprinter, 2024..

Focus: This Course focuses on Employability under CO1, CO2, CO3.

Outcome: After completion of course, the student will be able to:

- CO1: Implement Generative AI and LLMs models using Tensorflow and Pytorch.
- CO2: Design Chatbots using open-source APIs.
- CO3: Design, package and deploy GenAI microservices.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO2/PSO1
CO2	PO3,PO5/PSO2
CO3	PO3,PO5/PSO4